Identifying Human Settlements Using Deep Learning Models

A Comparative Assessment of Two Deep Leaning Models Using Landsat 8 and Sentinel 2 Imagery Services

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Human Settlement

Landsat Deep Learning Model (In Red)

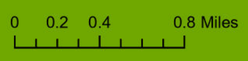
Kindu







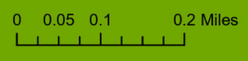
Lokandu





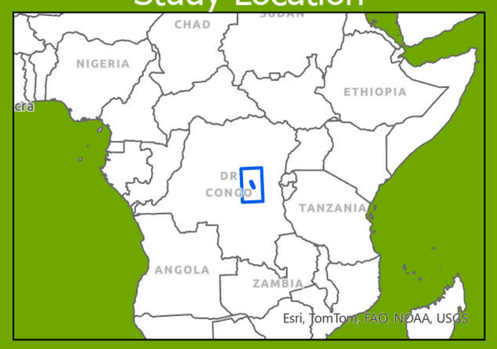


Rural Area





Study Location



1. What is deep learning and how do the two deep learning models work?
   a. Deep learningis part of a broader family of machine learning methods based on artificial neural networks with representation learning. Learning can be supervised semi-supervised or unsupervised.
   b. Deep learning models for human settlement detection analyze geospatial data, such as satellite imagery and other geographic datasets, to identify and map area where humans live and develop infrastructure. These models are designed to extract features thatdifferentiate between urban, rural, and uninhabited areas.
2. What is the acreage of human settlements for each study area?
   a. Kindu: 9068+ 2281 = 11349acres
   b. Lokandu: 195 + 38 = 233acres
   c. Rural Area: 13acres
3. How well do the models detect the extent of human settlements?
   a. Kindu: The Landsat model workedthe best here of all three settlementstudied. Reducing thepixel sizefrom 15 to 10increased the areas accounted for.
   b. Lokandu: The Landsat model workedthe best here of all three settlementstudied. Just like for Kindu, reducing thepixel sizefrom 15 to 10increased the areas accounted for.
   c. Rural Area: The Landsat model did not work for the rural area.
   d. Note: The Sentinel model did not work herefor any of the threehuman settlements after trying multiple settings forpixel size, tilessize, and padding.
4. Which deep learning model performs best at detecting human settlements?
   a. For this study, the Landsat model was the only one thatperformed at all. It would be interesting to see which one would be best if the Sentinel model worked.
5. What settings of the analysis help the model perform better?
   a. For the Landsat model,setting thepixel size smaller seemed to help itpick up more of the human settlementand other settings like padding and tile size did not seem to affect the results.
6. What are the suggestions for improvement or guidance for future users of this workflow?
   a. If someone has the time, try another computer if the results are not what is wanted.
   b. If someone does not have an available alternative computer, try uninstalling and reinstalling the software,possibly to a versionthat is known to work better.
   c. Be sure to set the processing template for the imagery layer to None. If this is not done, anerror occurs.
   d. Toggle thepixel size, tile size and padding andput these settings in the title tokeep track of which ones havebeen used.

Human Settlement/ Model	Pixel Size	Padding	Tile Size	Result
Kindu / Landsat8 - Success	10, 15, 30	100, 128	400	Result shownare for 15, 10 gives fewer and 30 desn't produce any more.
Lokandu / Landsat8 - Success	10, 15, 30	100, 128	400	Result shownare for 15, 10 gives fewer and 30 desn't produce any more.
Rural Area/ Landsat8 – Fail	None	None	None	Nothing worked
Kindu / Sentinel2 - Fail	None	None	None	Nothing worked
Lokandu / Sentinel2 - Fail	None	None	None	Nothing worked
Rural Area/ Sentinel2 - Fail	None	None	None	Nothing worked

Source: For imagery: Landsat 8 and Sentinel 2 imagery web services, For deep learning models: ArcGIS pretrained models